

RDU Hourly Temperature Forecast

Methods, leakage controls, model selection, and final results

Forecast task: issue one forecast at 2026-09-17 00:00 EDT for every hour through 2026-09-30 23:00 EDT (336 hours). The final test was not used for training, preprocessing fits, feature selection, tuning, or model selection.

Executive summary

The project replaces the original grid/reanalysis baseline with actual airport station reports from the Iowa Environmental Mesonet (IEM). It evaluates interpretable seasonal and anomaly models, required linear regression, regularized regression, gradient boosting, and random forest under the same historical forecast windows.

- Development data: 2015-01-01 through the forecast cutoff; RDU is the target, while GSO and CLT provide optional neighboring-station features.
- Clean data: 307,323 station-hours. Causal supervised data: 151,960 labeled rows across 453 historical forecast origins.
- Selection: 19 fixed configurations evaluated on eight 14-day historical forecast windows. The final test contains 336/336 observed hourly labels.
- Primary model selected before test: seasonal harmonic reference plus the previous 72-hour temperature anomaly, with the anomaly decaying over a 72-hour time constant.
- Primary test RMSE: 7.307°F (4.06°C); MAE: 6.097°F (3.39°C); R²: 0.492. The predefined nonlinear comparator scored RMSE 6.969°F but did not replace the primary model after test access.

Workflow at a glance

01 Acquire	02 Clean	03 Features	04-05 Validate/select	06 Freeze	07-08 Evaluate/explain
IEM routine reports	Hourly station labels	Past-only state + calendar	8 historical windows	Predictions + hashes	336 labels + plots

1. Data source and prediction target

The response variable is the hourly surface temperature reported at Raleigh-Durham International Airport (RDU). Data come from the public IEM ASOS/METAR archive. The same API supplies routine reports for RDU, Greensboro (GSO), and Charlotte (CLT). Variables are air temperature, dew point, sea-level pressure, wind speed, and wind direction.

- Development request: 2015-01-01 through, but never beyond, 2026-09-17 04:00 UTC.
- Final-test request: 2026-09-17 04:00 UTC through 2026-10-01 04:00 UTC, opened only after freezing.
- Only routine reports (report_type=3) are requested; high-frequency and special reports are not mixed into the target.
- Every download has a JSON sidecar containing the URL, bounds, download time, row count, observed range, and SHA-256 hash.

Hourly target definition

Routine reports occurring from minute :45 through :59 are assigned to the following nominal hour. If several reports map to the same station-hour, the latest is kept. The actual observation timestamp remains stored separately so feature availability can be audited. The target is therefore a station report near the end of the preceding hour, not an hourly average.

The retrospective archive does not preserve complete original issuance/revision timestamps. The pipeline therefore uses a 15-minute availability buffer as an explicit assumption. This limitation is documented rather than hidden.

2. Cleaning and quality control

- Parse timestamps as UTC and standardize station identifiers.
- Apply fixed, predeclared physical plausibility bounds: temperature -60 to 130°F; dew point -80 to 100°F; pressure 850 to 1100 hPa; wind 0 to 150 knots; direction 0 to 360 degrees.
- Convert out-of-range fields to missing, but do not manufacture target values. Missing target rows are excluded from fitting and scoring.
- Reject development labels or source observations at or after the cutoff. Recent RDU development coverage was 100% for the last 14 days.

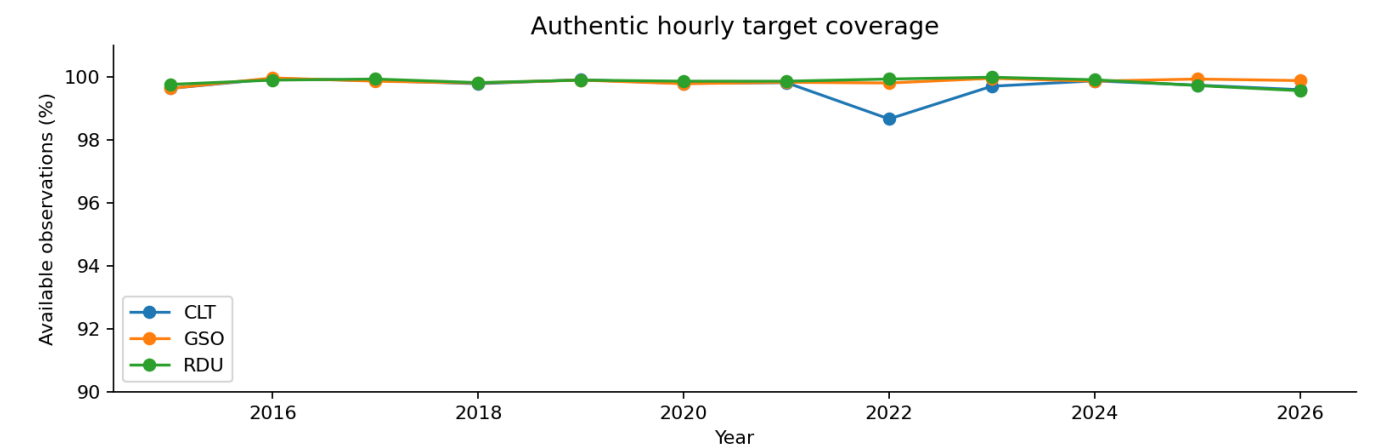


Figure 1. Authentic hourly target coverage by station and year.

3. Feature construction

Features are divided into groups so their contribution can be tested as a group rather than selected from the full dataset using the final test.

Feature group	Examples	Why it is available
Calendar	daily/yearly sine and cosine; second harmonics; daily × annual interactions	Future timestamps are known at issue time
Forecast lead	lead hour, log lead, smooth decay interactions	The 1-336 hour horizon is known
RDU state	24/72/168-hour anomaly, coverage, last anomaly and age, 24-hour variation/trend	Computed only from observations before the origin
RDU weather	dew-point depression, pressure mean/change, wind u/v	Past 24-48 hours only
Neighbors	GSO/CLT 24-hour anomalies and coverage	Past observations only

A causal annual reference is fitted separately for each simulated year using at most the preceding eight years and stopping before January 1 of that year. This prevents a historical 2020 feature from using a seasonal baseline trained on 2021-2026 labels.

For each weekly historical origin from 2018 onward, the code creates all 336 future rows. The feature values describing current weather are held at their origin-known values; only calendar and forecast-lead values change across the horizon. Actual weather observed during the forecast window is never used as a predictor.

Feature selection

Feature selection is validation-only paired ablation. With the same HGB settings and the same validation hours, the project compares state features, then adds local weather, then adds neighbors. For 15-leaf HGB, local weather reduced mean RMSE by 0.367°F; neighbors provided a further 0.030°F reduction and improved 6 of 8 folds. These modest gains justify retaining the neighbor version as the nonlinear comparator without claiming causal effects.

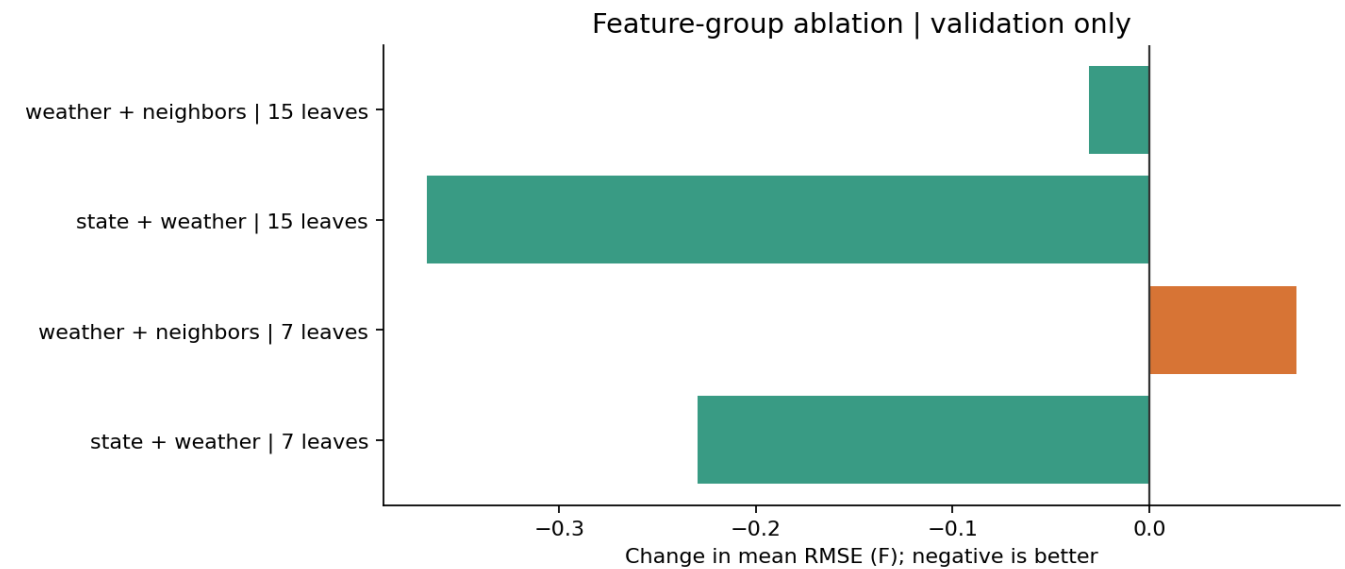


Figure 2. Validation-only feature-group ablation; negative change is better.

4. Models and selection

Family	Configurations	Interpretation
Baselines	seasonal climatology; repeat previous day	Minimum reference points
OLS	basic cycles; cycles + harmonics/interactions	Required transparent linear model
Ridge	alpha 10/1000; with/without trend	Regularized linear model
Anomaly	decay time 24/72/168 hours	Seasonal reference plus recent departure
Direct Ridge	state × horizon-decay interactions	Linear multi-step residual model
HGB	7/15 leaves × state/weather/neighbors	Small nonlinear residual trees
Random forest	100 trees; controlled depth/leaves	Tree ensemble comparator

Eight validation origins are used: September 17 in 2021, 2022, and 2023; August 15 and September 17 in 2024 and 2025; and August 15 in 2026. Each fold predicts a complete 336-hour window without receiving new observations.

Training rows are allowed only when their entire 336-hour target trajectory ends before the validation origin. Imputers and scalers are fitted inside each training fold. HGB early stopping is disabled so scikit-learn cannot create an uncontrolled random validation split.

Predeclared selection rule: find the minimum mean fold RMSE; among models within 1% of that value, select the least complex; use RMSE to break any remaining tie. OLS and the best nonlinear validation candidate are retained separately for comparison.

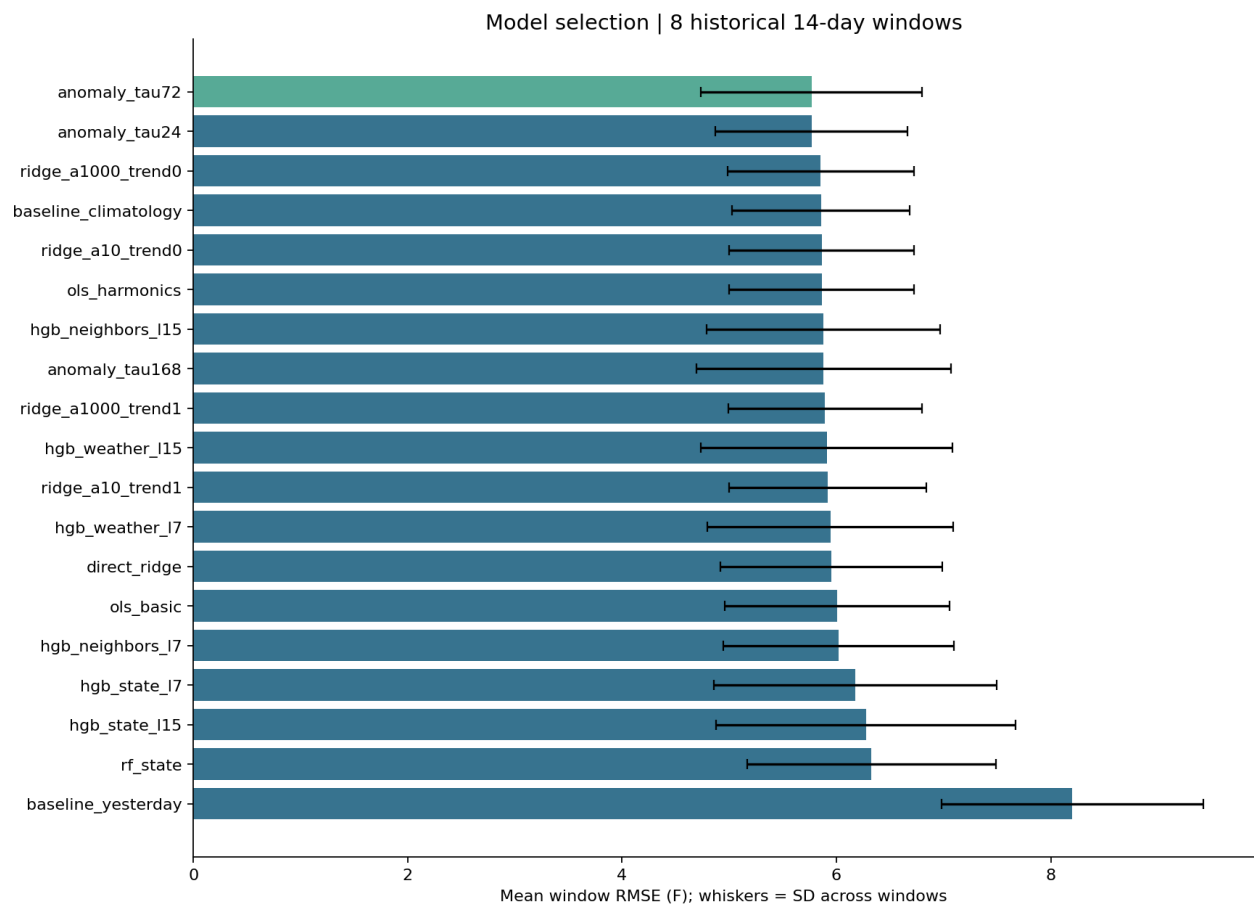


Figure 3. Mean RMSE over eight historical 14-day windows; whiskers show across-window SD.

5. Freeze and final-test isolation

The pipeline uses both procedural separation and file-level evidence. The purpose is to prevent an attractive final score from silently influencing the modeling process.

- Development scripts use only data/development; step 07 exclusively owns data/test_locked.
- API request guards reject development requests crossing the cutoff; loaded rows are checked again.
- Before test access, step 06 fits only the preselected final models, saves 336 predictions, models, features, coefficients and validation-residual interval rules.
- A freeze manifest records source, development data, model and prediction hashes. Step 07 refuses to run if any frozen item differs.
- Beginning test access creates a permanent lock. Development entry points then refuse to run.
- Final evaluation imports no training/model-selection function and records 0 fit calls, 0 feature-selection calls, and 0 model-selection calls.
- Nine synthetic tests verify strict cutoff behavior, request guards, locks, full-window purging, no label imputation, and invariance when future observations or training rows are poisoned.

6. How metrics are computed

Metric	Calculation	Meaning
RMSE	$\sqrt{\text{mean}((\text{prediction} - \text{actual})^2)}$	Penalizes large errors strongly; main selection metric
MAE	$\text{mean}(\text{abs}(\text{prediction} - \text{actual}))$	Typical absolute error in °F
Bias	$\text{mean}(\text{prediction} - \text{actual})$	Positive means forecasts are too warm on average
R ²	$1 - \text{SSE}/\text{SST}$	Fraction of test variation explained relative to its mean
Within ±2°C	share with absolute error ≤ 3.6°F	Descriptive tolerance rate added after scoring

These are regression metrics; R² and the tolerance rate are not classification accuracy. Missing labels would be excluded rather than interpolated, while coverage is reported. In this run all 336 labels are present.

7. Final results

Model	Role fixed before test	CV RMSE °F	Test RMSE °F	Test MAE °F	Test R²	Within ±2°C
anomaly_tau72	Selected primary	5.768	7.307	6.097	0.492	32.14%
ols_harmonics	Required OLS	5.863	7.653	6.398	0.443	31.85%
ridge_a1000_trend0	Best Ridge	5.854	7.669	6.423	0.440	31.25%
hgb_neighbors_l15	Nonlinear comparator	5.879	6.969	5.298	0.538	46.13%
baseline_climatology	Seasonal baseline	5.855	7.654	6.317	0.443	33.33%
baseline_yesterday	Persistence baseline	8.202	8.745	7.057	0.272	25.60%

The primary model was selected before final-test access. Its test RMSE is 7.307°F (4.06°C), MAE is 6.097°F (3.39°C), and R² is 0.492. Its mean bias is -0.272°F, indicating almost no average warm/cool offset, although hourly errors remain substantial.

The predefined HGB comparator achieved a lower test RMSE of 6.969°F and higher R² of 0.538. Because this became known only after opening the test, it is reported as a comparator rather than retrospectively promoted to the primary model. Future work may choose it using new development evidence, but must use another untouched final period.

The nominal historical 80% residual band covered only 64.58% of final observations. The interval is therefore not well calibrated for this weather period and must not be presented as guaranteed 80% coverage.

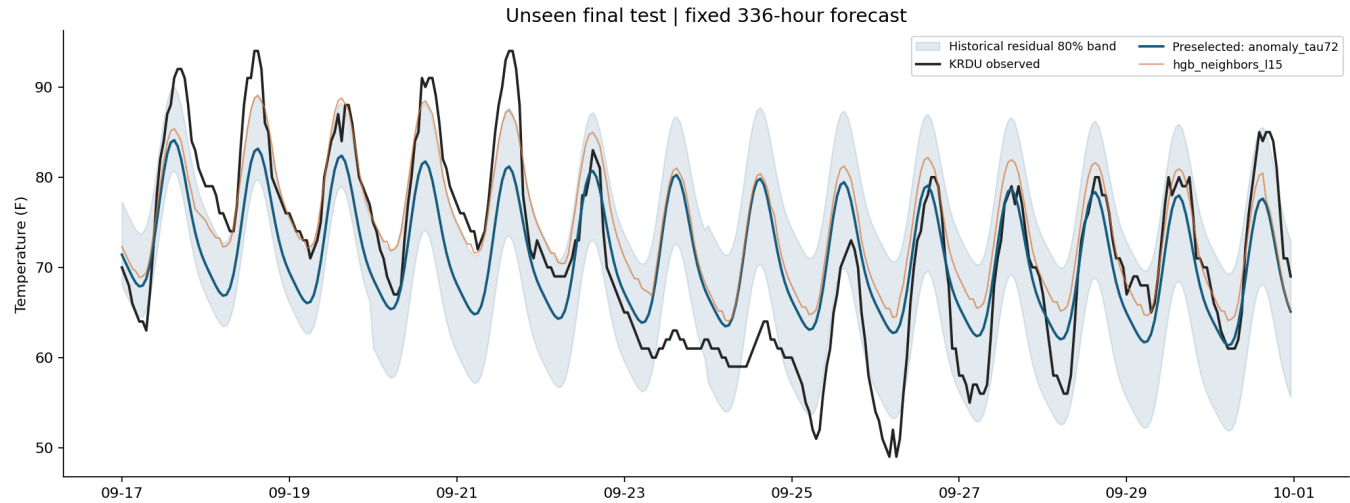


Figure 4. Frozen forecasts versus the unseen final observations.

8. Interpretation, limitations, and team handoff

Why the simple model was selected

On historical validation, anomaly_tau72 had the lowest mean RMSE (5.768°F) and is easy to explain: begin with the season/hour pattern, measure whether the last three days were warmer or cooler than normal, and let that adjustment fade. The HGB model was close on validation (5.879°F) but more complex. The fixed 1% rule therefore favored the anomaly model.

Limitations

- Station history alone cannot know future fronts, precipitation, clouds, or synoptic circulation across a 14-day horizon. This limits performance regardless of algorithm.
- Historical validation windows are few and were used for selection; their score is not an unbiased final estimate.
- The IEM archive may contain retrospective corrections, and the 15-minute availability buffer is an assumption.
- Feature effects and linear coefficients explain model behavior; they do not establish causal weather relationships.
- The final test is now published. Any model change needs a new untouched period before claiming independent performance.

What teammates should open

- Repository entry: README.md. Full methodology: rdu_forecast/README.md. Plain-language code map: rdu_forecast/CODE_GUIDE.md.
- Run python tools/09_verify_delivery.py inside rdu_forecast to verify evidence and recompute scores without training.
- Review reference_results/artifacts/results.html for all plots and tables. The branch is codex/rdu-station-forecast and is ready for a pull request into main.

References

Iowa Environmental Mesonet ASOS/METAR archive: <https://mesonet.agron.iastate.edu/request/download.phtml>
Scikit-learn lagged-feature example: https://scikit-learn.org/stable/auto_examples/applications/plot_time_series_lagged_features.html
Zeng et al., Are Transformers Effective for Time Series Forecasting?: <https://arxiv.org/abs/2205.13504>
Lou et al., Accurate Intelligible Models with Pairwise Interactions: <https://www.cs.cornell.edu/~yinlou/papers/lou-kdd13.pdf>